

An atomic Puiseux monoid of integer rank one

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Conjecture 00000002208

Claim addressed. Both versions of the source assert that a monoid generated by 1 and positive real numbers is atomic if and only if the integer rank of its generators is at least 2. The necessity direction is false.

Proposition 1. *The additive monoid $M = \langle 1, 2 \rangle \subseteq \mathbb{Q}_{\geq 0}$ is atomic, but the $\mathbb{Z}$ -module generated by its generators has rank 1.*

Proof. By definition,

$$M = \{a + 2b : a, b \in \mathbb{N}\} = \mathbb{N}.$$

Here $\mathbb{N}$ includes zero. Both specified generators are strictly positive; the generator 1 is present as required. This is also a Puiseux monoid under the usual convention of an additive submonoid of nonnegative rational numbers. Embedding $\mathbb{Q}$ in $\mathbb{R}$ gives exactly the same example under the source's positive-real convention.

The only additive unit of M is zero: if $a, b \in M$ and $a + b = 0$, nonnegativity implies $a = b = 0$. An atom in this reduced additive monoid is therefore a nonzero element u for which every decomposition $u = a + b$ with $a, b \in M$ has $a = 0$ or $b = 0$. The element 1 is an atom, since two natural numbers summing to 1 cannot both be positive. Every $n \in M$ is a finite sum of n copies of this atom. For $n = 0$ this is the empty sum. Thus M is atomic. In fact, any $n \geq 2$ decomposes as $1 + (n - 1)$ into two nonzero elements, so 1 is its only atom.

The integer span of the actual generators is

$$L = \text{span}_{\mathbb{Z}}\{1, 2\} = \mathbb{Z} \cdot 1 \subseteq \mathbb{Q},$$

because $2 = 2 \cdot 1$ and 1 is among the generators. The map $\mathbb{Z} \rightarrow L$, $k \mapsto k \cdot 1$, is a bijective $\mathbb{Z}$ -linear map: it is surjective by the description of L and injective because $1 \neq 0$ in $\mathbb{Q}$. Consequently L is a free module of rank 1, which is strictly less than 2. This proves the counterexample. $\square$

Formal correspondence. The accompanying Lean project defines M as the actual additive-submonoid closure of $\{1, 2\}$ in $\mathbb{Q}$, proves that its carrier consists exactly of natural-number casts, and proves the zero-unit property. Its atom definition uses decompositions by elements of this closure. The atomicity theorem supplies a finite indexed list of atoms whose sum is each given monoid element, including zero. The rank theorem concerns Mathlib's actual `Module.rank` of the actual integer span. It is obtained from an explicit linear equivalence with $\mathbb{Z}$, rather than an assigned numerical rank.

No restriction in either source excludes redundant positive generators or requires density. Even the singleton generating family $\{1\}$ would yield the same rank-one atomic monoid. This submission disproves the stated universal criterion; it makes no claim about a criterion with additional hypotheses.